USO DE GEMELOS DIGITALES PARA LA EFICIENCIA ENERGETICA

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Resumen

Como investigador novel, observo que los gemelos digitales (Digital Twins, DT) se han consolidado recientemente como una herramienta clave en la agricultura de precisión, al permitir la representación virtual y dinámica de sistemas agrícolas en tiempo real.

Estos modelos integran datos provenientes de sensores IoT, teledetección y maquinaria agrícola para simular procesos como crecimiento de cultivos, riego y fertilización, mejorando la toma de decisiones.

Los avances más recientes se centran en la integración de inteligencia artificial, aprendizaje profundo y modelos híbridos mecanísticos–data-driven, lo que incrementa la capacidad predictiva y habilita estrategias de optimización autónoma.

Asimismo, tecnologías emergentes como edge computing y aprendizaje federado permiten operar en tiempo real con mayor escalabilidad y privacidad.

Finalmente, los DT están evolucionando hacia ecosistemas interoperables que combinan simulación, control y ejecución en campo, aunque persisten desafíos asociados a la estandarización de datos, costos de implementación y complejidad computacional.

1. Introduction

Your introduction goes here! Simply start writing your document and use the Recompile button to view the updated PDF preview. Examples of commonly used commands and features are listed below, to help you get started.

1.1. Estado del arte en Mendoza

Once you're familiar with the editor, you can find various project settings in the Overleaf menu, accessed via the button in the very top left of the editor. To view tutorials, user guides, and further documentation, please visit our help library, or head to our plans page to choose your plan.

2. Materiales y métodos

2.1. How to create Sections and Subsections

2.1.1. Resultados de laboratorio de la UNCuyo

Simply use the section and subsection commands, as in this example document! With Overleaf, all the formatting and numbering is handled automatically according to the template you've chosen. If you're using the Visual Editor, you can also create new section and subsections via the buttons in the editor toolbar.

2.2. How to include Figures

First you have to upload the image file from your computer using the upload link in the file-tree menu. Then use the includegraphics command to include it in your document. Use the figure environment and the caption command to add a number and a caption to your figure. See the code for Figure 1 in this section for an example.

Figura 1: This frog was uploaded via the file-tree menu.

Note that your figure will automatically be placed in the most appropriate place for it, given the surrounding text and taking into account other figures or tables that may be close by. You can find out more about adding images to your documents in this help article on including images on Overleaf.

Producción de Textos e Hipertextos

Figura 2: TyHM Grupo JIGSA

- Archivo Readme.md
- Trabajo con tablas e Imágenes
- Construcción de Landing Page
- CV en LaTeX
- Uso de md en Google Docs
- Producción de textos html

Nombre	Apellido	Grupo	Whatsapp	Github
Gonzalo	Gonzalez	JIGSA	2625415602	https://github.com/gonzagonzalezgg-spec
Jael Gisel	Simonovich	JIGSA	2613431031	https://github.com/jaelsimonovich/TyHM_Modulo_1
Juan Ignacio	Ristagno	JIGSA	2613649714	https://github.com/Juan-Ignacio-Ristagno/Modulo-1-TyH
Sebastián	Egea	JIGSA	2616416126	https://segea86.github.io/TyHM_EGEA_Mod1/
Marcia Anahi	Benegas	JIGSA	2614727485	https://github.com/marciabenegas/-TyM-Modulo1

2.3. How to add Tables

Use the table and tabular environments for basic tables — see Table 1, for example. For more information, please see this help article on tables.

Item	Quantity
Widgets	42
Gadgets	13

Cuadro 1: An example table.

Ahora vamos a crear una tala de caracteres especiales

a	b	c	d
1	2	3	4
α	β	λ	μ
$A_1 \leq 30$	$A_{1B}^3 \geq 25$	$A_{1C} \approx 60$	$\lim_{x\to 0} A = \infty$

Cuadro 2: tabla de caracteres

2.4. How to add Comments and Track Changes

Comments can be added to your project by highlighting some text and clicking "Add comment" in the top right of the editor pane. To view existing comments, click on the Review menu in the toolbar above. To reply to a comment, click on the Reply button in the lower right corner of the comment. You can close the Review pane by clicking its name on the toolbar when you're done reviewing for the time being.

Track changes are available on all our premium plans, and can be toggled on or off using the option at the top of the Review pane. Track changes allow you to keep track of every change made to the document, along with the person making the change.

2.5. How to add Lists

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

... or bullet points ...

- Like this,
- and like this.

2.6. How to write Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

2.7. Como escribir ecuaciones de LATex

Hay dos formas de incluir ecuaicones en un texto con latex. La primer froma se utiliza para poner ecuaicones en el cuerpo del texto. La segunda forma es poner la ecuación como si fuese una imagen incrustada con el numero de ecuación y todo lo necesario par aque aparezca en el indice, incluido el caption.

2.7.1. Ecuaciones en linea

Segun Newton la expresión más conocida en la ingeniería para describir la relación entre las magnitudes masa y aceeleración es: $F = m * a$

2.7.2. Ecuaciónes en bloque

La segunda forma de poner las ecuaciones es introducirlas con la misma sintaxis del signo dolar $

$$e^{i\pi} + 1 = 0$$

Esta forma de escribir ecuaciones sirve tanto para Latex como para Markdown. Otra forma es utilizando el comando begin equation

$$e^{i\pi} + 1 = 0 \tag{1}$$

2.8. How to change the margins and paper size

Usually the template you're using will have the page margins and paper size set correctly for that use-case. For example, if you're using a journal article template provided by the journal publisher, that template will be formatted according to their requirements. In these cases, it's best not to alter the margins directly.

If however you're using a more general template, such as this one, and would like to alter the margins, a common way to do so is via the geometry package. You can find the geometry package loaded in the preamble at the top of this example file, and if you'd like to learn more about how to adjust the settings, please visit this help article on page size and margins.

2.9. How to change the document language and spell check settings

Overleaf supports many different languages, including multiple different languages within one document.

To configure the document language, simply edit the option provided to the babel package in the preamble at the top of this example project. To learn more about the different options, please visit this help article on international language support.

To change the spell check language, simply open the Overleaf menu at the top left of the editor window, scroll down to the spell check setting, and adjust accordingly.

2.10. How to add Citations and a References List

You can simply upload a `.bib` file containing your BibTeX entries, created with a tool such as JabRef. You can then cite entries from it, like this: [Gre93]. Just remember to specify a bibliography style, as well as the filename of the `.bib`. You can find a video tutorial here to learn more about BibTeX.

If you have an upgraded account, you can also import your Mendeley or Zotero library directly as a `.bib` file, via the upload menu in the file-tree.

2.11. Good luck!

We hope you find Overleaf useful, and do take a look at our help library for more tutorials and user guides! Please also let us know if you have any feedback using the **Contact us** link at the bottom of the Overleaf menu — or use the contact form at `https://www.overleaf.com/contact`.

Referencias

[Gre93] George D. Greenwade. The Comprehensive Tex Archive Network (CTAN). *TUGBoat*, 14(3):342–351, 1993.